THESIS PLAN

THESIS TITLE: A COMPARATIVE ANALYSIS OF HOW IN-HOUSE AND OUTSOURCED OPERATIONS AFFECT A COMPANY'S RISK PROFILE

Institution: Karelia University of Applied Sciences (Karelia UAS)
Research approach: Questionnaire-based comparative research
Status: Proposed plan, subject to review and approval

1. Background and Starting Point

Organisations must decide whether to perform activities using their own employees and resources or to obtain them from external service providers. These choices influence how organisations allocate resources, exercise control, manage dependencies, and respond to operational disruptions.

In-house operations may provide direct control over processes, employees, knowledge, and resources. However, they may also require substantial investment, internal expertise, and management capacity. Outsourced operations may provide access to specialised capabilities, external resources, and flexible capacity, but can introduce supplier dependencies, contractual risks, reduced direct control, and challenges in retaining organisational knowledge.

Neither operating model necessarily produces a lower overall risk. The resulting risk profile depends on the activity, organisational context, governance arrangements, capabilities, external dependencies, and measures used to prevent and manage disruptions.

We intend to investigate how in-house and outsourced operations affect operational and strategic business risks. The research will consider both the risks associated with each model and the arrangements through which organisations manage those risks.

2. Purpose and Aim

The purpose of this thesis is to compare how in-house and outsourced operations influence a company's risk profile from operational and strategic/business perspectives.

Our aim is to develop an evidence-based comparative framework that helps organisations understand how operating-model decisions affect risk exposure, control, dependencies, and resilience.

The research will examine whether particular risks are reduced, retained, transferred, introduced, amplified, or shared when activities are performed internally or outsourced.

3. Research Objectives

We will pursue the following objectives:

1. Identify the principal operational risks associated with in-house and outsourced operations.
2. Identify the strategic and business risks associated with each operating model.
3. Examine how control, resource allocation, organisational capabilities, supplier dependencies, and governance influence risk exposure.
4. Compare how organisations manage continuity, disruptions, and recovery under the two operating models.
5. Investigate the factors organisations consider when making make-or-buy decisions.
6. Develop a comparative risk framework based on the literature and collected questionnaire data.
7. Assess the framework's usefulness, limitations, and potential application in organisational decision-making.

4. Research Questions

The main research question is:

How do in-house and outsourced operations affect a company's operational and strategic/business risk profile?

The supporting research questions are:

- RQ1: What operational risks are associated with in-house and outsourced operations?
- RQ2: What strategic and business risks are associated with the two operating models?
- RQ3: How do control distribution, resource configuration, organisational capabilities, and external dependencies influence risk exposure?
- RQ4: How do organisations manage continuity, disruption, and recovery under in-house and outsourced arrangements?
- RQ5: What factors should organisations consider when evaluating the risk implications of make-or-buy decisions?

5. Knowledge Base and Theoretical Framework

We will establish the knowledge base through a literature review covering in-house operations, outsourcing, make-or-buy decisions, operational risk, strategic risk, governance, and organisational resilience.

The review will consider relevant theories where they help explain the research problem:

- Transaction Cost Economics: to examine transaction costs, uncertainty, contractual arrangements, and decisions about internal versus external provision.
- Resource-Based View: to examine internal capabilities, specialised knowledge, and the strategic value of resources.
- Resource Dependence Theory: to examine reliance on suppliers and other external organisations.
- Agency Theory and governance: to examine information asymmetry, monitoring, incentives, accountability, and control.
- Dynamic Capabilities: to examine how organisations adapt their resources and processes when conditions change.

We will assess each theory for its relevance to the research questions rather than include theories that do not materially contribute to the analysis.

The literature review will establish the concepts, evidence, and limitations needed to develop the comparative framework. We will verify sources and maintain a traceable relationship between cited evidence and the claims used in our analysis.

6. Conceptual Framework

We will use the following analytical sequence to organise the research:

Operating Model → Resource Configuration → Control Distribution → Dependency Structure → Risk Exposure → Risk Event or Disruption → Consequence → Response and Recovery → Residual Risk

This sequence will allow us to compare the two operating models without assuming that either is inherently safer.

We will examine the following operational risk dimensions:

- Process control and quality
- Reliability, downtime, and capacity
- Labour, skills, and maintenance

- Information flow and response time
- Flexibility and operational continuity
- Supplier performance and technology dependency
- Safety and compliance, where relevant to the collected evidence

We will examine the following strategic and business risk dimensions:

- Operating costs and transaction costs
- Capital requirements and financial exposure
- Strategic dependency and vendor lock-in
- Core competencies and knowledge retention
- Scalability and bargaining power
- Intellectual property and information control
- Business continuity and reputation

For each relevant risk, we will consider whether it is retained, transferred, introduced, reduced, amplified, or shared under the operating arrangement. We will distinguish risk transfer from actual risk reduction.

7. Research Approach and Design

We will conduct questionnaire-based comparative research supported by a literature review.

The questionnaire will collect information from professionals with relevant knowledge of operational management, procurement, supply chains, maintenance, quality, risk management, finance, outsourcing, or organisational decision-making.

The study will combine structured questions with optional open-ended questions. Structured responses will support descriptive comparisons, while open-ended responses will provide context about participants' experiences and risk-management practices.

The research will primarily examine reported experiences and perceptions. Unless the collected data supports stronger conclusions, we will not treat reported associations as proof that a particular operating model causes a specific risk outcome.

8. Research Population and Participants

Our intended participants are professionals who have direct experience of, or relevant decision-making responsibility for, in-house operations, outsourced activities, or both.

We will seek responses from different organisational roles and operating contexts where feasible. We will document the characteristics of the achieved sample sufficiently to explain which perspectives are represented.

Our initial outreach target is at least 250 qualified organisations or professionals. We aim to obtain at least 25 usable questionnaire responses, with a preferred target of 30–40.

These figures are planning targets, not guarantees of participation or statistical representativeness. If the final sample is small or unevenly distributed between operating models, we will limit the strength and scope of our comparisons accordingly.

9. Questionnaire and Data Collection

We will use a structured questionnaire organised around the following areas:

A. Company and respondent context
B. Operating model
C. In-house operations
D. Outsourced operations
E. Direct comparison
F. Continuity and resilience
G. Governance
H. Make-or-buy decisions
I. Open-ended question

We will use neutral wording and provide options such as “Not sure” and “Not applicable” where appropriate. We will avoid questions that unnecessarily request personal data or commercially sensitive information.

Before wider distribution, we will review and pilot the questionnaire to identify unclear wording, excessive completion time, missing response options, and other design problems. We will revise it where necessary.

We will collect data only after confirming any required research permits, approvals, and participant information arrangements with our thesis supervisor. We will document the data-collection method and relevant dates.

10. Data Management and Protection

We will follow our Data Management Plan throughout the project.

We intend to store research data on a password-protected personal computer with appropriate security measures and backups. We will confirm with our supervisor whether this arrangement is acceptable or whether Karelia UAS requires an approved university storage platform.

We will minimise the collection of personal data and protect information that could identify participants or organisations. If contact information is collected, we will store it separately from questionnaire responses. We will restrict access to research files to what is necessary for the project.

We will preserve original questionnaire exports, document data cleaning, use consistent file naming, and maintain traceability between the collected responses, analytical results, and thesis findings.

Our intended retention period is until thesis assessment is complete, followed by secure deletion where permitted. We will confirm the applicable institutional retention and disposal requirements before collecting data.

11. Research Ethics and Integrity

We will conduct the research in accordance with applicable Karelia UAS guidance on good scientific practice, research ethics, data protection, and responsible use of AI.

We will explain the study's purpose and participation arrangements to respondents before they take part. Participation will be voluntary, and we will handle responses according to the information provided to participants and the applicable data-protection requirements.

We will assess whether a research permit or other authorisation is required and obtain any necessary approval before data collection begins. We will also resolve any applicable consent, confidentiality, and data-processing requirements with our supervisor.

We will verify our references, attribute ideas appropriately, and distinguish source-supported claims from our own interpretations. We will not fabricate data, alter responses to support a preferred conclusion, or present AI-generated content as verified evidence.

As a two-person research team, we will agree on responsibilities for data collection, analysis, reference management, drafting, and review. Both researchers will review the accuracy and integrity of the final work.

12. Data Analysis

We will analyse the collected data in accordance with the research questions and the quality and size of the achieved sample.

For structured questionnaire responses, we may use frequencies, percentages, distributions, and cross-tabulations to describe participants' experiences and compare reported risks between operating models where the data permits.

We will examine open-ended responses through systematic coding and grouping of recurring themes. Relevant themes may include control, cost, supplier dependency, knowledge retention, continuity, governance, and recovery.

We will interpret the results alongside the literature review. We will identify areas of agreement, disagreement, and uncertainty between published evidence and participant responses.

We will document missing data, exclusions, coding decisions, and methodological limitations. We will not make claims about the wider population that cannot be supported by our sampling method and results.

13. Development of the Comparative Risk Framework

Using the literature and questionnaire findings, we will develop a comparative framework for assessing how in-house and outsourced operations affect a company's risk profile.

The framework will organise risks by operational or strategic/business dimension and consider:

- The operating model and activity being assessed.
- The source and nature of the risk.
- The organisation's control and dependency structure.
- Potential consequences and business exposure.
- Existing prevention, response, and recovery measures.
- The resulting or residual risk, where the evidence permits an assessment.

We will distinguish findings supported by the collected data from analytical interpretations and proposed framework elements.

The framework will be presented as a research output rather than as a universally validated risk-assessment instrument.

14. Validation and Limitations

We will assess the framework's consistency with the literature, traceability to the collected evidence, clarity of definitions, and usefulness in comparing operating models.

We will check whether the framework addresses the research questions and whether its categories can be applied consistently to the available evidence. If feasible and approved by our supervisor, we may seek feedback from relevant professionals or the supervisor on its clarity and practical relevance.

We recognise that the research may be limited by sample size, self-selection, differences between industries, participants' roles and experiences, incomplete responses, and reliance on self-reported information.

We will report these limitations and avoid claiming that the framework has been validated across all industries or organisational contexts unless further evidence supports that conclusion.

15. Research Schedule

We will follow the thesis schedule agreed with our supervisor and adjust it according to approval, recruitment, and data-collection progress.

Phase	Main activities
1. Planning	Confirm the topic, research questions, scope, and feasibility
2. Literature review	Identify, assess, and document relevant academic and professional sources
3. Research design	Finalise the conceptual framework, questionnaire, and analysis plan
4. Preparation and approval	Confirm permits, ethics, participant information, and data-storage arrangements
5. Pilot and collection	Pilot the questionnaire, revise it where necessary, and collect responses
6. Data preparation	Check, clean, code, and document the dataset
7. Analysis	Analyse structured and open-ended responses and compare the findings with the literature
8. Framework development	Construct and assess the comparative risk framework
9. Writing and review	Complete the thesis, verify references, and revise the findings and discussion

10. Finalisation	Complete integrity checks, apply Karelia UAS requirements, submit, and dispose of data according to the approved plan

We will update the schedule when actual progress, recruitment, or supervisor feedback requires changes.

16. Roles and Responsibilities

We will work as a team and jointly take responsibility for the research design, ethical conduct, evidence quality, interpretation, and final thesis.

We will agree on the division of practical tasks, including literature searching, questionnaire administration, data cleaning, analysis, drafting, and proofreading. Task allocation will not remove either researcher's responsibility to understand and review the overall research.

We will consult our supervisor when we encounter uncertainty about the research scope, methodology, permissions, data protection, analysis, or institutional requirements.

17. Expected Results and Contribution

We expect the research to produce:

1. A structured overview of operational and strategic/business risks associated with in-house and outsourced operations.
2. A comparison of how control, capabilities, dependencies, governance, and continuity arrangements influence risk exposure.
3. An evidence-based comparative risk framework for supporting make-or-buy decisions.
4. Practical considerations that organisations can use when reviewing operating arrangements and associated risks.

The findings will depend on the evidence collected and the limitations of the research design. We will not assume in advance that one operating model is preferable to the other.

18. Expected Thesis Structure

We plan to organise the final thesis in accordance with the applicable Karelia UAS thesis-reporting instructions:

1. Introduction
2. Knowledge Base

3. Methods
4. Implementation
5. Results
6. Discussion
7. References
8. Appendices

The final structure and formatting will be confirmed against the current instructions and our supervisor's guidance.

19. Feasibility and Approval

Before moving into full data collection, we will confirm with our supervisor that:

- The research questions and scope are sufficiently clear.
- The literature review provides an adequate knowledge base.
- The questionnaire and proposed analysis are suitable for answering the research questions.
- The participant recruitment approach and response targets are realistic.
- Any required research permits and ethical arrangements have been resolved.
- The Data Management Plan and storage arrangements are acceptable.
- The research schedule and division of responsibilities are feasible.

We will incorporate feedback from the thesis-plan review and obtain the necessary approvals before proceeding with activities that require them.

20. Overall Research Logic

Our research will follow this sequence:

Research Problem → Literature Review → Conceptual Framework → Questionnaire Design → Required Approvals → Data Collection → Data Quality Checks → Analysis → Comparative Risk Framework → Validation and Limitations → Thesis Writing → Final Integrity Review → Submission

This sequence will help us maintain alignment between the research questions, evidence, analysis, and conclusions while ensuring that research ethics, data management, and integrity are considered throughout the project.